Project-1 : Text to Image

```
# !pip install diffusers --upgrade
```

```
!pip install invisible_watermark transformers accelerate safetensors
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.1.0)
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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch->i
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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
```

```
!nvidia-smi
```

```
Mon Apr  6 19:43:52 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.82.07              Driver Version: 580.82.07      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   37C    P8             12W /   70W |       0MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
```

```
|=============================================================================================|
|  No running processes found                                                                 |
+---------------------------------------------------------------------------------------------+
```

```
from diffusers import DiffusionPipeline
import torch

pipe = DiffusionPipeline.from_pretrained("stabilityai/stable-diffusion-xl-base-1.0", torch_dtype=torch.float16, use_safetens
pipe.to("cuda")

# if using torch < 2.0
# pipe.enable_xformers_memory_efficient_attention()

prompt = "dog is sitting wearing black specs "

images = pipe(prompt=prompt).images[0]
```

```
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning yo
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning yo
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Loading pipeline components...: 100%  7/7 [00:07<00:00, 1.28s/it]
Loading weights: 100%  196/196 [00:00<00:00, 342.09it/s, Materializing param=text_model.final_layer_norm.weight]
Loading weights: 100%  517/517 [00:02<00:00, 224.86it/s, Materializing param=text_projection.weight]
100%  50/50 [00:37<00:00, 1.28it/s]
/usr/local/lib/python3.12/dist-packages/diffusers/pipelines/stable_diffusion_xl/pipeline_stable_diffusion_xl.py:748: FutureW
  deprecate(
```

```
!pip install gradio
```

```
Requirement already satisfied: gradio in /usr/local/lib/python3.12/dist-packages (5.50.0)
Requirement already satisfied: aiofiles<25.0,>=22.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (24.1.0)
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Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0
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Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer<1.
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer<
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3
```

```
images
```

```
import torch
from diffusers import DiffusionPipeline
import gradio as gr
# Load Stable Diffusion XL model
pipe = DiffusionPipeline.from_pretrained(
    "stabilityai/stable-diffusion-xl-base-1.0",
    torch_dtype=torch.float16,
    use_safetensors=True,
    variant="fp16"
)
pipe.to("cuda")
# Image generation function
def generate_image(prompt):
    image = pipe(prompt=prompt).images[0]
    return image
# Gradio Interface
demo = gr.Interface(
    fn=generate_image,
    inputs=gr.Textbox(label="Enter your prompt"),
    outputs=gr.Image(label="Generated Image"),
```

```
    title="Stable Diffusion XL Image Generator",
    description="Generate AI images using Stable Diffusion XL"
)
demo.launch()
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
Loading pipeline components...: 100%  7/7 [00:03<00:00, 2.21it/s]
Loading weights: 100%  196/196 [00:00<00:00, 358.63it/s, Materializing param=text_model.final_layer_norm.weight]
Loading weights: 100%  517/517 [00:01<00:00, 633.25it/s, Materializing param=text_projection.weight]
It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting `share

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://36ea181be7ad2a668f.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the
```

Stable Diffusion XL Image Generator

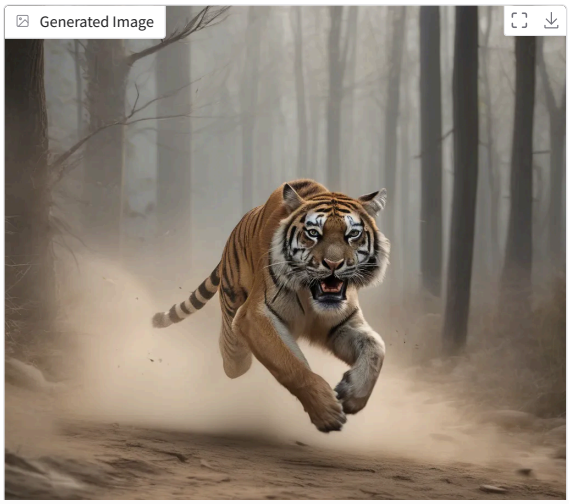
Generate AI images using Stable Diffusion XL

Enter your prompt

Tiger running behind the deer

Clear Submit

Generated Image



Next steps: Deploy to Cloud Run

```
!nvidia-smi
```

```
Mon Apr  6 19:47:27 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.82.07              Driver Version: 580.82.07      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   62C    P0             30W /   70W |    8543MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A           17540      C   /usr/bin/python3                       8540MiB |
+-----------------------------------------------------------------------------------------+
```

```
!nvidia-smi
```

```
Mon Apr  6 19:47:27 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.82.07              Driver Version: 580.82.07      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   62C    P0             30W /   70W |    8543MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
```

```
+-----------------------------------------+-----------------------+----------------------+

+---------------------------------------------------------------------------------------+
| Processes:                                                                            |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=======================================================================================|
|    0   N/A  N/A           17540      C   /usr/bin/python3                       8540MiB |
+---------------------------------------------------------------------------------------+
```